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He^+ Ion Irradiation Effects on the Phase Stability and Microstructure of High-Purity Zr_3AlC_2

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Abstract

High-purity Zr_3AlC_2 samples (>92 wt%) were synthesized and irradiated at room temperature using 100 keV He^+ ions at fluences of 2×10^{16} and 1×10^{17} ions/cm^2. As a Zr-based MAX phase, Zr_3AlC_2 is a promising candidate for accident-tolerant fuel cladding due to its compatibility with Zr alloys and the low neutron absorption cross-section of Zr. Our results show that irradiation induces a decrease in the a-lattice parameter and an increase in the c-lattice parameter, along with the formation of anti-site defects and decomposition into ZrC. Cracks preferentially appear along (1000) planes. These findings suggest that Zr_3AlC_2 has limited structural stability under low-temperature helium irradiation.

Keywords: MAX phases; spark plasma sintering; ion irradiation; Zr-based ceramics; Zr_3AlC_2

1. Introduction

MAX phases are a class of ternary layered compounds with the general formula $M_{n+1}AX_n$, where M represents an early transition metal, A is an A-group element, and X is carbon or nitrogen [1,2]. These materials typically exhibit a hexagonal close-packed (hcp) structure and possess a unique combination of metallic and ceramic properties, including high thermal and electrical conductivity, excellent mechanical strength, corrosion resistance, and oxidation resistance [3,4]. Some MAX phases also demonstrate favorable irradiation tolerance, which has attracted significant interest for their potential applications in nuclear environments [5,6].

Among various M elements (such as Ti, Cr, and V), zirconium (Zr) is particularly notable for its low neutron absorption cross-section. ZrC-based ceramics have already shown promising performance under reactor conditions, making Zr-based MAX phases a subject of increasing attention [7]. In 2016, Lapauw et al. [8] successfully synthesized Zr_3AlC_2, which has been proposed as a potential Accident Tolerant Fuel (ATF) coating material due to its ability to prolong fuel life and enhance accident resistance. However, the synthesis of high-purity Zr_3AlC_2 remains challenging. Many research groups have attempted various sintering approaches, yet the resulting MAX phase content is typically below 70 wt% [9,10]. To improve phase purity, some researchers have explored doping strategies. For instance, Zapata-Solvas et al. introduced Si and Ti into the precursor powders and synthesized $Zr_3(Al_x,Si_{1-x})C_2$ [11] and $(Zr_{1-x}Ti_x)_3AlC_2$ [12]; similar approaches were also reported by Lapauw et al. [13].

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To date, irradiation studies of MAX phases have primarily focused on Ti-, Cr-, and V-based systems, while investigations on Zr-based MAX phases remain limited. Bowden et al. [14] examined the effect of 2 MeV proton irradiation on Zr_3AlC_2 and found that it exhibits excellent defect recovery capabilities above 400 °C. They also reported lattice parameter changes under irradiation, characterized by a decrease in the *a*-lattice constant and an increase in the *c*-lattice constant. Qarra et al. [15] studied the effects of 22 MeV Au^+ ions on Zr_2AlC at different temperatures and observed that at room temperature, Zr_2AlC becomes largely amorphous after irradiation, whereas it retains structural stability above 350 °C. In another study, Qarra [16] further investigated the structural response of $Zr_3(Al_{0.9}Si_{0.1})C_2$ under 52 MeV I^{9+} irradiation at room temperature, finding that this material displayed poor irradiation tolerance, with phase transformations from α-$Zr_3(Al_{0.9}Si_{0.1})C_2$ to β-$Zr_3(Al_{0.9}Si_{0.1})C_2$ and subsequent decomposition into ZrC and $Zr(Si/Al)_2$.

Helium is a significant fission product generated during the transmutation of actinides. High-energy He ions can directly displace atoms via elastic collisions, introducing structural damage [17]. Additionally, implanted He tends to migrate and agglomerate, leading to the formation of He bubbles that cause significant volumetric swelling and embrittlement [18–21]. Therefore, He-induced damage is a critical concern for nuclear structural materials. To date, there have been no reports on the effects of He ion irradiation on Zr_3AlC_2.

In addition, previous studies have predicted that Zr_3AlC_2 exhibits better chemical compatibility with zirconium-based alloys compared to many other MAX phases, making it particularly attractive for integration with existing cladding materials in nuclear reactors [22].

In this study, high-purity Zr_3AlC_2 samples (>92 wt%) were synthesized and irradiated with 100 keV He^+ ions at room temperature. Two fluences were applied: 2×10^{16} ions/cm^2 and 1×10^{17} ions/cm^2, corresponding to approximately 1.3 dpa and 6.6 dpa, respectively. After irradiation, a significant increase in ZrC content was observed in the irradiated region, accompanied by the disappearance of the layered MAX phase structure. At a fluence of 1×10^{17} ions/cm^2, cracks were observed on the sample surface, primarily occurring within individual grains.

2. Experimental Procedure

High-purity Zr_3AlC_2 was synthesized by spark plasma sintering (SPS) combined with a powder-stacking technique [23]. The starting materials included ZrH_2 (>99% purity), ZrC (>99% purity, particle size < 3 μm), and Al (>99% purity, particle size < 45 μm). ZrH_2 was prepared in-house by heating metallic zirconium at 350 °C for 2 h under a hydrogen atmosphere, while ZrC and Al powders were purchased from Nilaco Corporation. A powder mixture with a molar ratio of ZrH_2:ZrC:Al = 2:3:2 was loaded into the bottom of a graphite die, and a layer of pure ZrC powder was subsequently placed on top of the mixed powder. SPS was carried out at 1450 °C for 1 h. During sintering, a liquid phase L(Zr–Al) formed in the lower mixed-powder region and migrated upward, where it reacted with the upper ZrC layer to form Zr_3AlC_2. After sintering, the consolidated bulk could be clearly divided into two regions. The upper region exhibited a high Zr_3AlC_2 purity of approximately 93 wt%, whereas the lower region contained a lower Zr_3AlC_2 fraction of about 47 wt%. A distinct interface was observed between the two regions, and high-purity Zr_3AlC_2 could be obtained by mechanically cutting along this interface. Part of the material was ground into powder for X-ray diffraction (XRD) analysis, while the remaining bulk specimens were used for irradiation experiments.

For irradiation, the bulk Zr_3AlC_2 samples were cut and polished into thin plates with dimensions of approximately 1 mm × 7 mm × 7 mm, followed by final polishing using a

colloidal silica suspension. Helium ion irradiation was carried out at Hokkaido University using a 200 keV ion implanter. The He^+ ion energy was 100 keV, and the fluences were 2×10^{16} and 1×10^{17} ions/cm^2. The damage profiles in Zr_3AlC_2 were simulated using the SRIM 2013 software package (Figure 1), employing standard displacement threshold energies of 25 eV for Zr and Al, and 28 eV for C, with the density of Zr_3AlC_2 set to 5.62 g/cm^3. The maximum damage depth was located at approximately 530 nm. The fluences of 1×10^{17} and 2×10^{16} ions/cm^2 corresponded to peak damage levels of approximately 6.6 and 1.3 dpa, respectively.

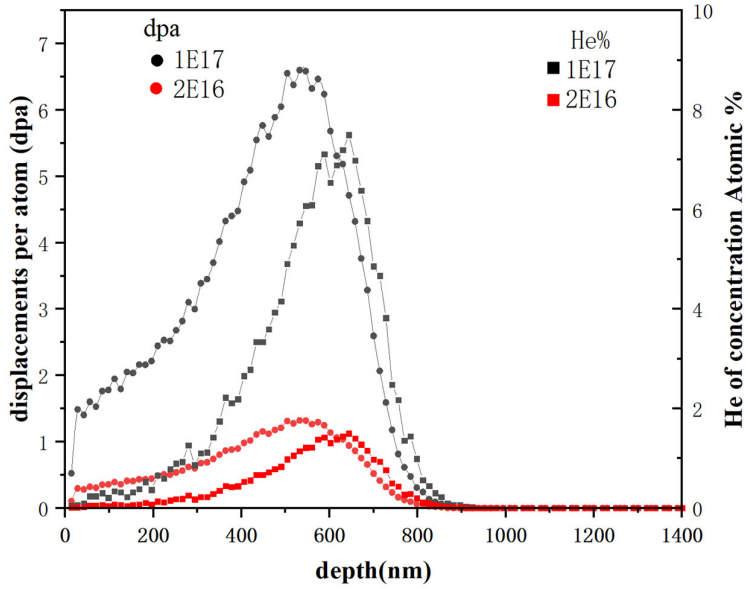

Figure 1. SRIM simulation results showing the damage profiles for 100 keV He^+ ions at fluences of 1×10^{17} and 2×10^{16} ions/cm^2.

The phase composition of the powdered samples was characterized using X-ray diffraction (XRD, Rigaku SmartLab, Tokyo, Japan) with Cu Kα radiation (λ = 1.5406 Å). The diffraction patterns were collected at room temperature using a grazing-incidence geometry with an incident angle of 3° and a scanning rate of 1°/min.

Rietveld refinement of the XRD data was performed using the RIETAN-2000 software (RIETAN-FP 3.12). The initial structural models for all identified phases were obtained from standard crystallographic databases in the form of crystallographic information files (CIFs).

The instrument resolution function (IRF) was experimentally determined using a Si powder measured under identical instrumental and geometric conditions. The obtained IRF was incorporated into the refinement to correct for instrumental broadening effects, enabling a more reliable determination of phase composition.

To investigate the surface structure of the irradiated samples, grazing incidence X-ray diffraction (GIXRD, Rigaku SmartLab) was employed with an incident angle of 0.5° and a scanning rate of 0.01°/min. The surface morphology before and after irradiation was examined using a scanning electron microscope (SEM, JEOL JSM-6510LA, JEOL Ltd., Tokyo, Japan). High-resolution scanning transmission electron microscopy (JEM-ARM200F NEOARM, JEM-ARM200F NEOARM, Tokyo, Japan) was used to observe the atomic arrangement of the samples, and energy-dispersive X-ray spectroscopy (EDS) equipped on the instrument was utilized to analyze the elemental distribution. The specimens for STEM observation were prepared by focused ion beam (FIB, FB-2100) milling.

3. Results

Figure 2A shows the Rietveld refinement of the XRD pattern collected from powdered Zr_3AlC_2 obtained by crushing the high-purity bulk sample prior to irradiation. The observed diffraction pattern is well reproduced by the calculated profile, indicating a satisfactory refinement quality. The diffraction peaks are predominantly indexed to the Zr_3AlC_2 phase, with minor contributions from ZrC and $ZrAl_2$. Figure 2B presents the quantitative phase analysis obtained from the Rietveld refinement. The content of Zr_3AlC_2 in the upper region exceeds 92 wt%.

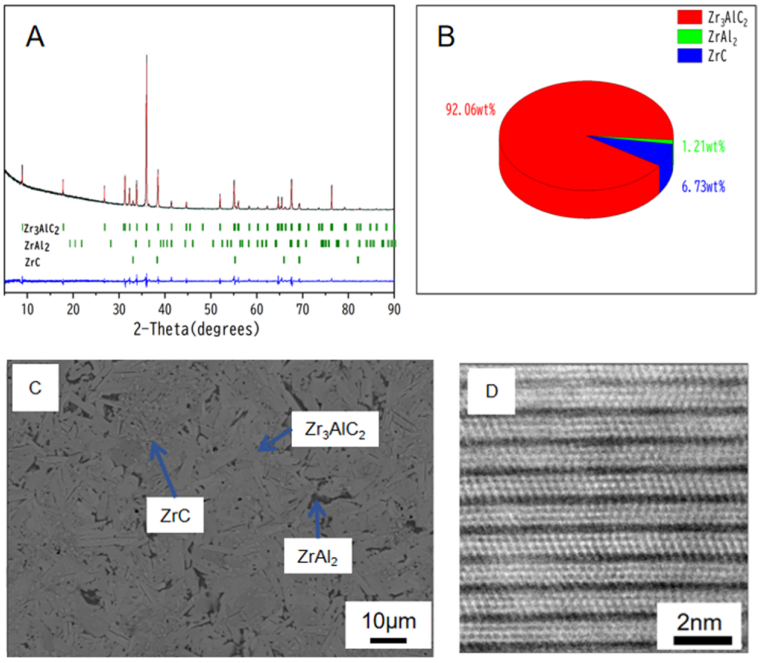

Figure 2. (**A**) XRD pattern of the powdered sample. (**B**) Phase composition and corresponding lattice parameters. (**C**) SEM surface image of the bulk sample. (**D**) STEM image of Zr_3AlC_2.

During SPS, a significant amount of liquid-phase $ZrAl_2$ diffused upward from the mixed powder region into the ZrC layer. $ZrAl_2$ reacts with the top-layer ZrC to form Zr_3AlC_2, and with continued holding at high temperature, the excess $ZrAl_2$ gradually migrates toward the outer edge of the die, leaving only a small amount of residual $ZrAl_2$ in the final product. This reaction–diffusion mechanism ensures the formation of a high-purity Zr_3AlC_2 layer in the upper region of the bulk sample.

The sintered samples exhibit two distinct regions. As the Zr_3AlC_2 fraction in the lower region is relatively low, the detailed SEM results of this region are presented in Supplementary Figure S1, while the following discussion focuses on the upper region.

Figure 2C displays a scanning electron microscopy (SEM) image of the bulk sample. The microstructure is dominated by elongated, bar-like Zr_3AlC_2 grains, with $ZrAl_2$ mainly located at the grain boundaries. Additionally, a small number of granular ZrC particles are observed. Figure 2D shows a scanning transmission electron microscopy (STEM) image taken along the [11–20] zone axis. The typical layered structure of the MAX phase is clearly visible. The brighter contrast corresponds to Zr atomic columns, while the darker contrast indicates the positions of Al atoms.

Figure 3 presents the XRD pattern of bulk Zr_3AlC_2 prior to irradiation (incident angle: 5°, scanning rate: 2° min^{-1}) and the grazing-incidence XRD (GIXRD) patterns of the irradiated samples (incident angle: 0.5°, scanning rate: 0.1° min^{-1}). New diffraction peaks are observed in the irradiated region, which can be indexed to the $(ZrAl_{1/3})_3C_2$ phase. The

intensities of these peaks increase with increasing irradiation dose, indicating a progressive irradiation-induced phase evolution within the near-surface region.

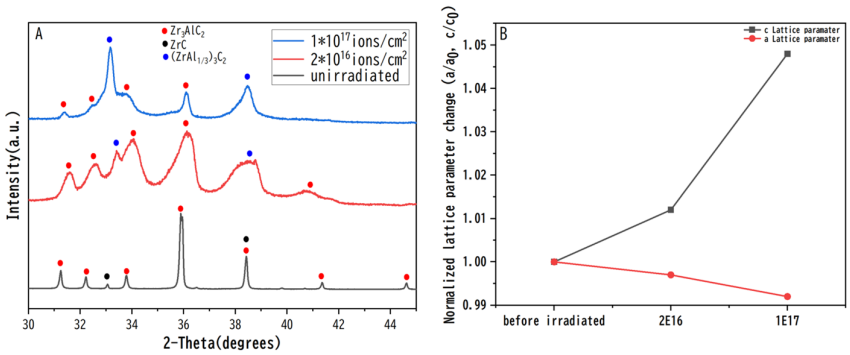

Figure 3. (**A**) GIXRD patterns of Zr_3AlC_2 samples before and after He^+ ion irradiation measured at an incident angle of 0.5°. (**B**) Normalized lattice parameter changes (a/a_0 and c/c_0) of Zr_3AlC_2 as a function of irradiation fluence, where a_0 and c_0 are the lattice parameters of the unirradiated sample.

In Zr_3AlC_2, the bonding between Zr and Al layers is relatively weaker compared with the strong Zr–C bonds. Ion irradiation is therefore expected to preferentially disrupt the Al atomic layers, which is likely to result in the formation of a high density of Zr–Al antisite defects. The accumulation of such defects is believed to drive the structural evolution toward the so-called β-MAX phase. Similar irradiation-induced phase transformations associated with antisite defect formation have been widely reported in Ti_3AlC_2 [19], Cr_2AlC [24,25], and high-entropy MAX-phase materials [26].

Another prominent feature is the pronounced shift in the residual Zr_3AlC_2 diffraction peaks after irradiation, indicating irradiation-induced changes in the lattice parameters. Figure 3B summarizes the variations in the lattice constants a and c. The lattice parameter a exhibits a slight decrease, whereas the lattice parameter c increases significantly, revealing a clear anisotropic lattice response. Such behavior is consistent with previous observations reported for He^+-irradiated Ti_3AlC_2 [27,28].

The observed expansion of the lattice parameter c can be reasonably interpreted in light of simulation results reported by Ling [29], who suggested that irradiation-induced *c*-axis expansion in Al-containing MAX phases mainly originates from two factors: (i) the formation of a high density of M–A antisite defects, which alters the charge density distribution, and (ii) the displacement of A atoms into interstitial positions under ion impacts. By contrast, the conversion of C atoms into interstitials induced by ion irradiation has been reported to result in a slight contraction of the lattice parameter a, in agreement with the experimental trend observed in this study.

In addition, significant peak broadening of Zr_3AlC_2 is observed after irradiation, indicating the formation of a high density of defects within the irradiated region. However, no amorphization is detected, suggesting that Zr_3AlC_2 exhibits good resistance to irradiation-induced amorphization.

Figure 4a shows a STEM image of the unirradiated Zr_3AlC_2 sample observed along the [11–20] zone axis. The pristine Zr_3AlC_2 exhibits a well-defined layered structure, where every three Zr atomic layers are separated by a single Al atomic layer. The Al atoms appear in the darker contrast regions, while the Zr atoms correspond to the brighter regions.

Figure 4b shows a STEM image of the maximum damage region in the sample irradiated with 1×10^{17} ions/cm^2 of 100 keV He^+ ions. After irradiation, the layered structure of Zr_3AlC_2 disappears, and Zr and Al atoms are uniformly distributed throughout the crystal. This structural change is attributed to the formation of numerous anti-site defects (Zr_Al and Al_Zr), which cause atomic rearrangement.

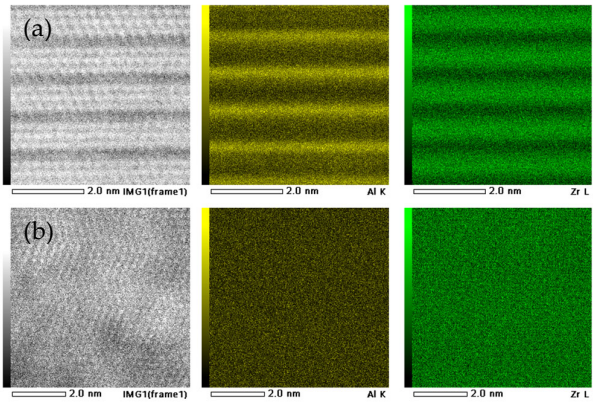

Figure 4. (**a**). STEM image of unirradiated Zr_3AlC_2 observed along the [11–20] zone axis. (**b**). STEM image of Zr_3AlC_2 irradiated with a fluence of 1×10^{17} ions/cm^2, observed along the [11–20] zone axis.

Figure 5 illustrates the stacking sequence of Zr_3AlC_2 before and after irradiation. Prior to irradiation, the stacking sequence of Zr_3AlC_2 can be described as . . .a–b–c–Al–c–b–a–Al–a–b–c. . ., where adjacent Zr–C layers separated by Al layers exhibit a symmetric distribution. After helium ion irradiation, Zr and Al atoms become indistinguishable, yet the crystal structure still maintains a symmetric arrangement. This suggests that He ion irradiation induces atomic rearrangement in Zr_3AlC_2 but does not completely destroy its crystalline structure. Such structural changes correspond to the enhanced (103) diffraction peak observed in the XRD results.

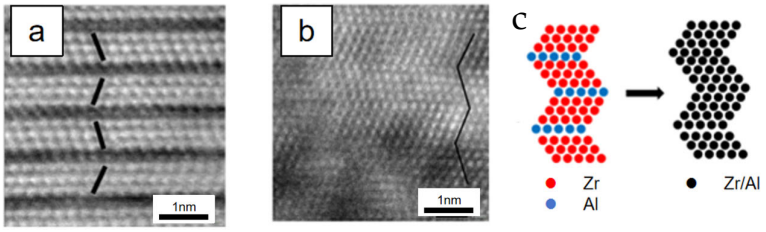

Figure 5. Stacking sequences of the samples before and after irradiation. (**a**) Crystal structure of the sample before irradiation, (**b**) Crystal structure of the sample after irradiation, (**c**) Schematic diagram of structural changes of the sample before and after irradiation.

A similar phenomenon was reported by Yang et al. [19], who irradiated Ti_3AlC_2 with 50 keV He ions at room temperature. They concluded that the irradiation induced a structural transformation from Ti_3AlC_2 to β-Ti_3AlC_2. By analogy, the post-irradiation structure of Zr_3AlC_2 can be represented by the chemical formula $(ZrAl_{0.33})_3C_2$.

This irradiation-induced degradation of the layered structure in MAX phases has also been reported in irradiation studies of other MAX-phase materials, such as Ti_3AlC_2 [30] and Ti_3SiC_2 [31]. It is widely accepted that this structural degradation is primarily associated with the formation of M–A antisite defects, in which M and A atoms partially occupy each other's original lattice sites, thereby disrupting the characteristic layered ordering.

Figure 6A,B present STEM images of the samples irradiated with He ions at fluences of 2×10^{16} ions·cm^{-2} and 1×10^{17} ions·cm^{-2}, respectively. The observations were performed along the [0001] zone axis. The bright contrast features observed in the images are attributed to He bubbles, which is consistent with previous STEM observations reported for He-ion-irradiated MAX-phase materials.

The average size of He bubbles at the two irradiation fluences was quantified using ImageJ software (1.54p) based on statistical analysis of multiple regions. With increasing irradiation fluence, the average He bubble diameter increases from 0.74 nm to 1.43 nm, indicating a clear fluence-dependent bubble growth behavior.

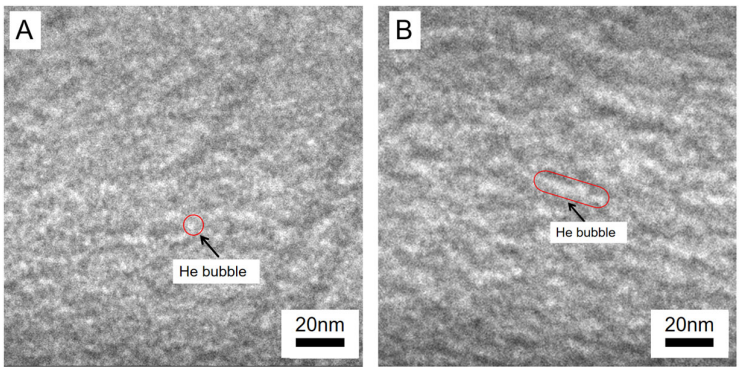

Figure 6. Cross-sectional TEM images of Zr_3AlC_2 samples irradiated with He ions along the [0001] direction: (**A**) sample irradiated to a fluence of 2×10^{16} ions·cm^{-2}; (**B**) sample irradiated to a fluence of 1×10^{17} ions·cm^{-2}.

As shown in Figure 6B, when the irradiation fluence increases to 1×10^{17} ions·cm^{-2}, partial coalescence of adjacent He bubbles is observed, leading to the formation of elongated, band-like, large He bubbles. A similar bubble coalescence behavior has been reported by Zhang et al. [32] in their study on He-ion-irradiated Ti_3SiC_2. Furthermore, Pang et al. [33] reported that during He-ion irradiation of Ti_3AlC_2, He atoms preferentially accumulate in the Al layers, which can promote *c*-axis lattice expansion and potentially increase the susceptibility of MAX phases to irradiation-induced cracking.

Figure 7a,b show SEM images of the sample before irradiation and after irradiation at a fluence of 2×10^{16} ions/cm^2, respectively. No cracks are observed on the sample surface either before irradiation or under this relatively low irradiation dose. Figure 7c presents the SEM image of the sample irradiated at a fluence of 1×10^{17} ions/cm^2, where distinct cracks can be seen within the red-circled region. Figure 7d shows that the cracks occur within the grains rather than along the grain boundaries, which is consistent with observations reported for various MAX-phase materials, such as Ti_2AlC [34], Ti_3AlC_2 [32], and $Zr_3(Al_{0.9}Si_{0.1})C_2$ [16].

A focused ion beam (FIB) was used to prepare a TEM lamella from the cracked region. Figure 7e reveals that the crack penetrates through the entire irradiated zone and extends into the underlying unirradiated region, terminating in a spherical damage zone at its tip.

Figure 7f presents a selected area electron diffraction (SAED) pattern from the cracked region, showing that the crack preferentially propagates along the (100) crystallographic plane. The cracking behavior of Zr_3AlC_2 under helium ion irradiation is similar to that observed in Ti_3AlC_2. Shen et al. [20] reported that irradiation of Ti_3AlC_2 with 50 keV He^+ ions at room temperature also led to crack formation, with cracks predominantly growing along the (100) planes.

It is generally believed that MAX phases can develop cracks under irradiation through two main mechanisms. The first is the generation of internal stresses due to anisotropic changes in lattice parameters caused by irradiation. The second involves the accumulation of radiation-induced defects, which tend to concentrate near the Al atomic layers located on the (100) planes. This defect localization promotes preferential crack formation along the (100) planes.

Irradiation induces pronounced structural changes in Zr_3AlC_2. Owing to the relatively weak bonding between Zr and Al atoms, the Al atomic layers are preferentially disrupted under irradiation, leading to the formation of a high density of Zr–Al antisite defects. The accumulation of these defects is believed to give rise to two concurrent effects.

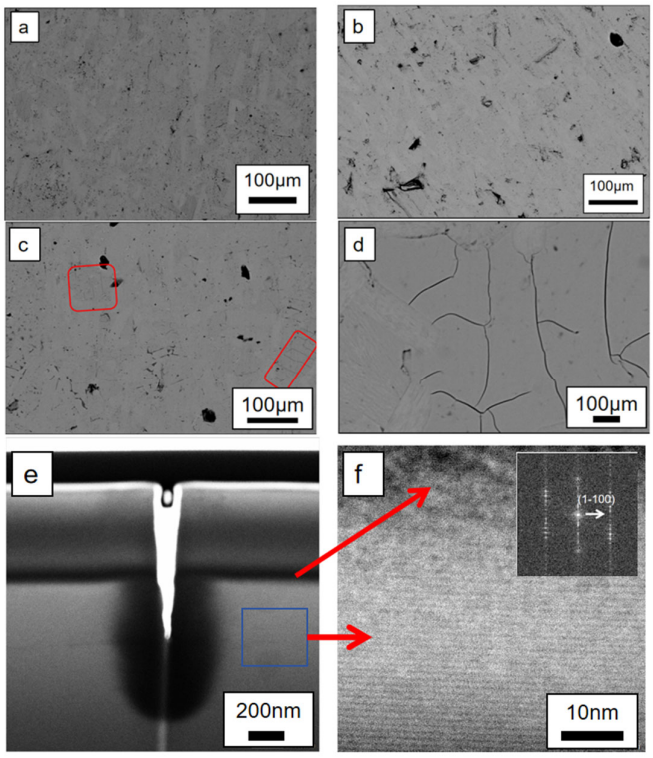

Figure 7. (**a**) Surface SEM image of the unirradiated sample. (**b**) Surface SEM image of the sample irradiated with a fluence of 2×10^{16} ions/cm^2. (**c**) Surface SEM image of the sample irradiated with a fluence of 1×10^{17} ions/cm^2; the red box indicates the crack region. (**d**) High-magnification surface SEM image of the sample irradiated with a fluence of 1×10^{17} ions/cm^2. (**e**) TEM image of the crack. (**f**) High-magnification STEM image of the crack region and SAED pattern of the unirradiated region.

First, the original α-MAX phase (Zr_3AlC_2) partially transforms into the β-MAX phase, which can be described as $(ZrAl_{1/3})_3C_2$. Second, the residual Zr_3AlC_2 phase undergoes pronounced lattice distortion, characterized by a slight contraction of the *a*-axis lattice parameter and a significant expansion of the *c*-axis lattice parameter.

In addition, Pang et al. [33] reported that during He-ion irradiation of Zr_3AlC_2, He bubbles preferentially accumulate within the Al atomic layers, which further enhances the expansion of the *c* lattice parameter. The combined effects of defect accumulation, phase transformation, lattice distortion, and He-bubble-induced *c*-axis expansion generate substantial internal stresses within the material, ultimately promoting the initiation and propagation of irradiation-induced cracks.

In summary, high-purity Zr_3AlC_2 samples (>92 wt%) were successfully synthesized using a spark plasma sintering (SPS) method combined with a powder stacking technique. Helium ion irradiation experiments conducted at room temperature revealed that Zr_3AlC_2 exhibits limited irradiation resistance under low-temperature conditions. Irradiation induced a phase transformation from α-Zr_3AlC_2 to β-Zr_3AlC_2 and partial decomposition into ZrC. The lattice parameters were also affected by irradiation, with a decrease in the *a*-lattice constant and an increase in the *c*-lattice constant. At a fluence of 1×10^{17} ions/cm^2, transgranular cracks appeared on the sample surface, preferentially propagating along the (100) crystallographic planes.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/ceramics9020023/s1, Figure S1: SEM of samples from the lower layer region.

Author Contributions: Conceptualization, Y.W.; methodology, Y.W.; software, Y.W.; validation, Y.W.; formal analysis, Y.W.; investigation, Y.W.; resources, Y.W.; data curation, Y.W.; writing—original draft preparation, Y.W.; writing—review and editing, N.H., H.O. and S.I.; visualization, Y.W.; supervision, N.H.; project administration, Y.W.; funding acquisition, N.H. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: The authors declare no conflict of interest.

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